

ARGUS: Auditing Causal Identification Credibility in Environmental Policy Evaluation

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Abstract. Empirical evaluations of environmental policy lean heavily on difference-in-differences (DID) designs whose credibility rests on identification assumptions that are often asserted rather than audited. We propose **ARGUS**, a responsible-AI framework that audits the *identification credibility* of such studies rather than adjudicating the true causal effect, which is never observed. ARGUS decomposes identification into auditable dimensions, gathers in-paper evidence, assesses each assumption–implication–evidence chain, and localizes risk in a transparent, human-reviewable report. We validate the auditor with *synthetic flaw injection*, which provides reproducible local ground truth without the true effect. A deterministic keyword baseline shows that, once the leakage that makes naive evaluation circular is removed, detection collapses to a low rate and the baseline is blind to flawed-but-present evidence—quantifying the gap this doctoral research aims to close.

Keywords: Responsible AI · Causal inference · Difference-in-differences · Environmental policy · LLM agents.

1 Introduction

Difference-in-differences (DID) is the workhorse design for evaluating environmental policy. Studies of carbon pricing, emissions-trading pilots, and industrial-emission standards routinely compare treated and untreated units before and after a policy to recover its causal effect. The credibility of any such estimate rests entirely on a chain of *identification assumptions*: that treated and control units would have moved in parallel absent the policy, that units did not anticipate it, that staggered adoption is handled with an appropriate estimator, that there is no interference between treated and control units, and so on. A decade of methodological work has shown how fragile each of these assumptions can be, and how easily a standard two-way fixed-effects specification can mislead under heterogeneous and staggered treatment timing [5,3,7,4,6].

Yet in applied work these assumptions are frequently *asserted* rather than *audited*. A paper states that parallel trends “hold”, shows an event-study plot, and moves on; a reader who wants to know whether the identification argument is actually supported by the reported evidence must reconstruct it by hand, dimension by dimension. This is slow, expert-intensive, and inconsistent across

reviewers. The result is a reliability gap in a body of evidence that increasingly informs environmental policy.

Our stance. We argue that identification *credibility* is auditable even though the causal *effect* is not. In a DID design the counterfactual is never observed, so it is hopeless—and, we contend, the wrong target—to ask an automated system “is the estimated effect true?”. Instead we ask a question that can be grounded in the paper itself: *is each identification assumption supported by the evidence the study reports, and where is that support weakest?* This reframing turns an unanswerable causal question into a tractable auditing problem over assumptions, their testable implications, and the in-paper evidence that bears on them. Figure 1 illustrates the view: credibility varies across assumptions, evidence, and policy context, and a useful system should localize where it is fragile rather than issue a verdict.

ARGUS. We introduce **ARGUS**, a responsible-AI framework that audits the identification credibility of DID environmental-policy studies. ARGUS decomposes identification into a fixed set of auditable dimensions, gathers the supporting evidence each dimension requires, assesses every assumption–implication–evidence chain, localizes risk across dimensions, and produces a transparent, evidence-anchored report for a human expert. It is deliberately a *bounded-agentic* system: the pipeline between modules is fixed and deterministic, and model-driven agency is confined to two scoped sub-tasks. Because the true effect is unavailable as ground truth, we validate the auditor with *synthetic flaw injection*: we perturb a credible paper with one known identification flaw and check whether ARGUS detects it, stays quiet on the clean version, and localizes the right dimension. The injected flaw is local ground truth that requires no access to the true effect.

Contributions. This paper, presented as a doctoral-research plan, makes four contributions. (1) **ARGUS**, a bounded-agentic framework that audits identification credibility—not causal truth—in DID environmental-policy evaluation (§3). (2) An *identification rubric* of eleven auditable dimensions and a *flaw taxonomy* mapping each known threat one-to-one onto a dimension. (3) A *flaw-injection evaluation methodology* that yields reproducible local ground truth—detection, false alarm, and localization—without the true effect (§4). (4) An honest deterministic baseline and a *negative result*: once the leakage that makes naive evaluation circular is removed, keyword-based detection collapses from a trivial 1.000 to 0.182, and the baseline is systematically blind to threats where flawed-but-plausible evidence is present (§5). This quantifies the reasoning gap that the remainder of the doctoral programme is designed to close.

2 Related Work

DID identification and its threats. A large recent econometrics literature has formalized when DID estimators fail. Goodman-Bacon [5] decomposes the two-

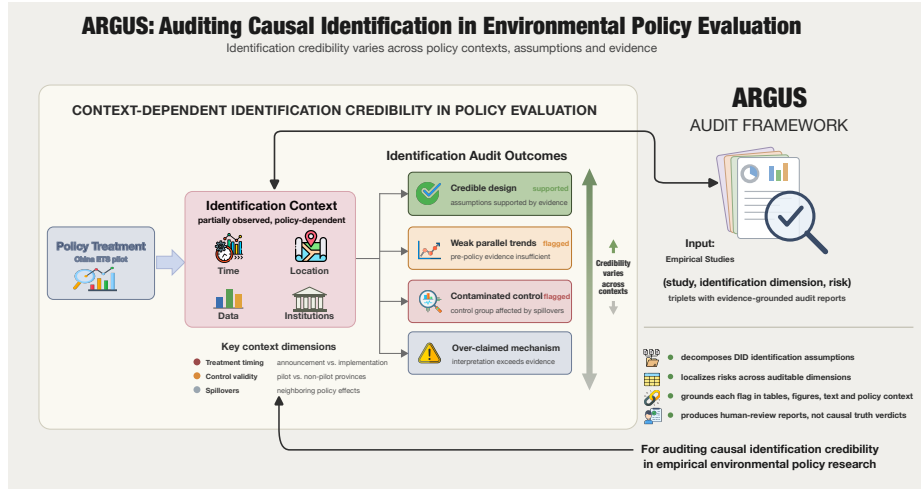


Fig. 1. Context-dependent identification credibility in empirical environmental policy evaluation. Causal-identification risks vary across treatment timing, control-group validity, spillovers, concurrent policies and other contexts, which ARGUS audits by decomposing assumptions and localizing evidence-based risk flags.

way fixed-effects estimator under variation in treatment timing and exposes the “forbidden” comparisons it makes between newly and already-treated units; de Chaisemartin and D’Haultfœuille [4] show such estimators can place negative weights on treatment effects; and Callaway and Sant’Anna [3] and Sun and Abraham [7] propose estimators robust to heterogeneous, staggered effects. Roth et al. [6] survey the design’s assumptions and the subtleties of pre-trend testing. ARGUS does not add an estimator; it *operationalizes* this literature as an auditable rubric, turning each known threat into a dimension whose supporting evidence can be checked in a given paper (§3).

Responsible AI for empirical research. A growing body of work applies AI to research reliability—automated reproducibility checking, statistical-reporting validation, and AI-assisted peer review. ARGUS is complementary but differs in target and in epistemic posture. First, it audits *identification credibility* specifically, the part of an empirical argument that determines whether a causal claim is licensed at all, rather than surface statistical reporting. Second, and centrally, it does not issue verdicts: it produces evidence-anchored, human-reviewable reports that localize risk, leaving adjudication to a domain expert. This human-in-the-loop, “flag-not-judge” stance is a deliberate response to the fact that the counterfactual is unobserved and that automated causal verdicts would be both unreliable and unaccountable.[2,1]

Bounded LLM agents. Tool-using agents in the ReAct style interleave reasoning and acting over an open horizon [8]. ARGUS deliberately *bounds* this paradigm.

Agency is confined to two scoped sub-tasks (extraction and assessment), each run as a fixed-tool, hard-step-budget loop with a logged trace, while the inter-module control flow is ordinary deterministic code. We trade autonomy for two properties the framework depends on: *reproducibility*, so that the flaw-injection metrics are stable and comparable across runs, and *evidence-traceability*, so that every flag a human reviews is anchored to inspectable evidence. We view this restraint as a feature for responsible auditing rather than a limitation.

3 The ARGUS Framework

3.1 Identification rubric

ARGUS decomposes a DID study into eleven auditable dimensions: parallel trends, no anticipation, staggered-timing handling, SUTVA/spillovers, control-group construction, specification, inference, sample-period selection, concurrent policies, placebo/robustness, and data measurement. Each dimension is expressed as an *assumption* (what must hold for identification), a testable *implication* (what should therefore be observable), and the *evidence* a credible paper would report to support it—for example, the parallel-trends dimension expects an event-study or pre-trend test with null pre-period coefficients. The rubric is declarative, stored in a configuration file (`identification_dimensions.yaml`), so that the set of audited dimensions is explicit and inspectable rather than buried in code. A companion flaw taxonomy (§4) maps each known identification threat one-to-one onto a dimension, which both documents the rubric’s coverage and supplies the perturbations used for evaluation.

3.2 Bounded-agentic pipeline

The audit runs a fixed, deterministic sequence—*decomposition* $\rightarrow$ *extraction* $\rightarrow$ *assessment* $\rightarrow$ *localization* $\rightarrow$ *report*—shown in the top band of Figure 2. The stage order is written in code and is never chosen by a model. *Decomposition* instantiates the rubric for the paper at hand. *Extraction* gathers, per dimension, the in-paper evidence relevant to that assumption. *Assessment* evaluates each assumption–implication–evidence chain and emits a per-dimension risk judgement with a rationale and cited evidence. *Localization* aggregates these judgements into a risk map that ranks where support is weakest. *Report* renders a transparent, evidence-anchored document for a human reviewer.

Agency is confined to the two stages where open-ended judgement actually helps, extraction and assessment. Each invokes a bounded ReAct-style loop with three constraints: (i) a *small fixed tool set* passed in per call—`evidence_search` over the paper’s text, figures and tables, `figure_parse` for structured figure/table content, and `policy_lookup` for policy metadata such as announcement versus implementation dates; (ii) a *hard step budget* (`max_steps`) so a run cannot wander indefinitely; and (iii) a *logged trace* of every step, tool call, and cited piece of evidence. All other stages are plain orchestration with no agent loop.

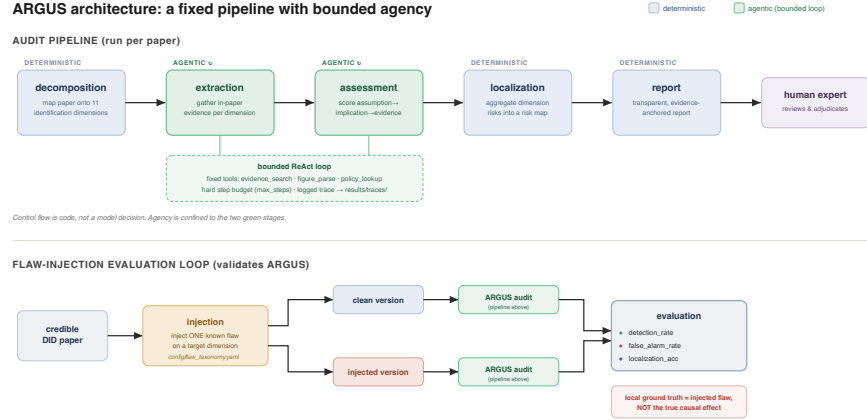


Fig. 2. ARGUS architecture. *Top:* the audit pipeline is a fixed deterministic sequence; agency is confined to the two green stages, each a bounded ReAct loop with a fixed tool set, a hard step budget, and a logged trace. *Bottom:* the flaw-injection evaluation loop injects one known flaw, audits the clean and injected versions, and scores detection, false alarm, and localization against the injected flaw as local ground truth—never against the unobserved true effect.

Why bound the agency. A fully autonomous agent would make per-run behaviour path-dependent and high-variance, which would undermine the central evaluation claim of §4—that flaw injection provides *reproducible* local ground truth. Restricting agency to two clearly scoped sub-tasks preserves reproducibility (stable, comparable metrics) and evidence-traceability (every judgement anchored to logged evidence a human can inspect). The expert sits at the end of the pipeline: ARGUS surfaces and localizes identification risk, it does not adjudicate it.

3.3 Output

ARGUS emits (*study, identification dimension, risk*) triplets, each with an evidence-grounded rationale, together with the human-readable report. It never produces a verdict on the true causal effect. This output shape is what makes the system both auditable by a human and measurable by the evaluation that follows.

4 Evaluation by Flaw Injection

4.1 Local ground truth without the counterfactual

The defining difficulty in evaluating a causal-identification auditor is that the quantity of ultimate interest, the true causal effect, is never observed. We therefore do not attempt to certify that ARGUS recovers true effects. Instead we

construct *local* ground truth by controlled perturbation. Beginning from a credible paper, we inject exactly one known flaw that targets a single identification dimension, drawn from the flaw taxonomy (`flaw_taxonomy.yaml`). Because we control which dimension we damaged, we know the correct audit outcome for that dimension—without ever needing the true effect. The bottom band of Figure 2 shows the loop: the clean and injected versions are each audited, and three metrics are computed.

- **Detection:** does the injected version raise risk on the targeted dimension above a threshold?
- **False alarm:** does the clean version stay quiet on that dimension?
- **Localization:** is the targeted dimension the one the auditor flags most strongly?

4.2 Realistic perturbation, not sentinels

How a flaw is injected is methodologically load-bearing. A naïve injector appends a sentence that *names* the flaw—for instance, “treated and control have non-zero pre-period coefficients with significant leads”. If the detector then searches for exactly those phrases, detection becomes a tautology: the injector plants the very string the detector greps for. In our initial implementation this produced a perfect-looking but meaningless detection rate of 1.000.

We replace this with *realistic structural perturbation* of two kinds. *Omission* flaws *remove* the section, figure, or table that supplies the support for a dimension—for example, deleting the placebo-robustness section or the balance table—modelling a paper that simply omits the check; an auditor can flag this non-circularly by noticing the *absence* of evidence. *Commission* flaws *rewrite* a section in flawed-but-natural prose, the way a real flawed paper would describe a mistaken design choice (e.g. pooling all province-years in a two-way fixed-effects regression under staggered timing), *without* containing any of the phrases the detector looks for. We enforce the latter with a regression test that fails if an injected paper contains the target dimension’s negative-signal phrases, so the circularity cannot silently return. Detection is thereby a genuine task rather than a string match, which is what makes the results in §5 meaningful.

5 Preliminary Results

As a lower bound and a sanity check on the evaluation methodology, we implement a deterministic, non-LLM baseline: extraction retrieves evidence by token overlap, and assessment scores each dimension from the presence or absence of supporting and risk-indicating vocabulary. We run it over all eleven flaws on two parsed-paper fixtures—a generic supported DID study and a China carbon-ETS pilot fixture—injecting one flaw per run. Table 1 reports the headline metrics; the full per-flaw breakdown is in `experiments/phase1_pilot/results.md`, and Figure 3 is generated directly from the live evaluation.

Table 1. Honest baseline under realistic flaw injection (`max_steps = 1`, eleven flaws, one injected per run). Removing the sentinel leakage that made evaluation circular drops detection from 1.000.

Fixture	detection false alarm localization		
<code>clean_supported</code>	0.182	0.000	0.182
<code>china_carbon_ets</code>	0.091	0.000	0.091
(circular, pre-fix)	1.000	0.000	1.000

What the baseline gets right. The baseline never raises a false alarm: on the clean papers it stays quiet on every dimension (false-alarm rate 0.000). It also correctly flags the two omission flaws—`measurement_break` and `spillover_contamination`—whose vocabulary does not appear elsewhere in the paper, detecting them by the *absence* of evidence. This is exactly the behaviour we want from a lower bound and confirms the evaluation harness is measuring something real.

Where it fails, and why that matters. Beyond those two cases, detection is low (0.182 and 0.091). Two failure modes explain it (Figure 3, panel B). First, *cross-section vocabulary bleed*: deleting the placebo section does not help when another section also mentions “placebo”, so retrieval still returns a hit and the baseline judges the dimension supported. Second, and more fundamentally, the baseline is blind to *every* commission flaw, where flawed-but-plausible evidence is present and uses the expected topic vocabulary. The lesson is sharp: keyword presence is not evidence *adequacy*. Distinguishing a sound parallel-trends argument from a superficially similar but flawed one requires reasoning about what the evidence actually shows, not whether the right words appear. This is precisely the gap a reasoning-based auditor must close, and it frames the doctoral programme that follows (§6).

6 Discussion and Research Plan

Threats to validity. The present results are deliberately preliminary. They use a small number of synthetic fixtures rather than a real corpus, and the baseline is an intentional lower bound, so its low detection rate should be read as a calibrated starting point, not as evidence about the eventual system. More fundamentally, flaw injection certifies an auditor against the *injected* threat on a single dimension; it does not certify the absence of other, un-injected threats, and it inherits whatever blind spots the flaw taxonomy has. We therefore treat detection and localization as necessary, not sufficient, evidence of an auditor’s quality, and pair them with the false-alarm rate to guard against a system that flags everything.

Research plan. The doctoral programme is organized around closing the reasoning gap that §5 makes concrete, in four stages. (1) *Reasoning-based extraction*

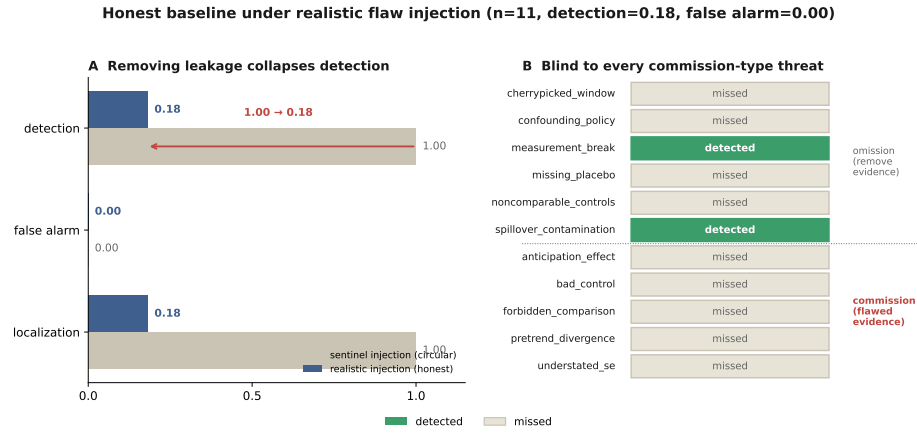


Fig. 3. Honest baseline under realistic flaw injection. (A) Removing the sentinel leakage that made evaluation circular collapses detection from 1.000 to 0.182, while false alarm stays at 0. (B) Per-flaw outcomes grouped by injection type: the baseline catches only omission-type threats whose vocabulary is not echoed elsewhere, and is blind to every commission-type threat where flawed-but-plausible evidence is present. Generated from the live evaluation by `experiments/phase1_pilot/make_figure2.py`.

and assessment. Replace the keyword policy inside the bounded loop with a model that judges evidence *adequacy*—does the reported event study actually support parallel trends?—evaluated against the baseline under the same flaw-injection protocol, so that any gain is attributable and reproducible. (2) *A real annotated corpus.* Build a corpus of published DID environmental-policy studies with expert annotations of identification risk per dimension, moving evaluation beyond synthetic fixtures and enabling agreement studies against human auditors. (3) *Human-expert studies.* Measure whether ARGUS reports actually help domain experts—calibration of trust, time saved, and errors avoided—since the system is explicitly designed to assist, not replace, expert judgement. (4) *Broader policy domains.* Extend beyond the China emissions-trading case to other environmental and, eventually, general empirical-economics settings, testing how far the rubric-plus-injection methodology generalizes.

7 Conclusion

We have argued that the right target for AI assistance in empirical policy research is not the unobservable causal effect but the *identification credibility* of the study that estimates it. ARGUS operationalizes this stance: a bounded-agentic pipeline decomposes identification into auditable dimensions, gathers and assesses in-paper evidence, and localizes risk in a transparent, human-reviewable report, while confining model agency to two scoped, traceable sub-tasks. We validate such auditors with synthetic flaw injection, which supplies reproducible local

ground truth without access to the true effect, and we show that once the leakage making this evaluation circular is removed, a keyword baseline is revealed to be insufficient—blind to flawed-but-plausible evidence. That negative result is the point of departure for the doctoral programme: reasoning over evidence adequacy, validated on a real annotated corpus and with human experts in the loop. We believe auditing identification credibility is both a tractable and a responsible role for AI in empirical environmental-policy research.

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